Predictors of sPCI–pPCI Discordance: Robust Linear Model (RLM) Analysis

Date: April 16, 2026
Source: `notebooks/02-assoc_difference_ols_robust.ipynb`

---

1. Background and Objective

Following the OLS analysis (see `02-assoc_difference_report.md`), which identified violations of normality and homoskedasticity, this analysis re-examines the predictors of sPCI–pPCI discordance using **Robust Linear Regression (RLM)** via iteratively reweighted least squares (IRLS) with a Huber-T M-estimator. This approach down-weights influential outliers and provides more reliable inference under non-normal, heavy-tailed residuals.

---

2. Study Sample

N = 412 patients after excluding tumour type 8.
Tumour type 1 (n = 122) serves as the reference category.

Outcome: `raw_diff` = sPCI − pPCI (signed discordance)

---

3. RLM Regression Results

Model: IRLS, Huber-T norm, scale estimated via MAD, covariance type H1
No. Observations: 412 | **Df Residuals:** 400 | **Df Model:** 11 | **Iterations:** 26

Regression Coefficients

Predictor	Coefficient	SE (asymptotic)	SE (bootstrap)	z	p	95% CI (bootstrap)
Intercept	+7.69	2.195	2.168	3.51	**< 0.001**	[+3.66, +12.09]
Sex (male)	−0.83	0.635	0.654	−1.30	0.193	[−2.09, +0.41]
Age	−0.030	0.024	0.025	−1.27	0.206	[−0.079, +0.020]
Zn HIPEC	+2.54	1.487	1.206	1.71	0.088	[−0.24, +4.45]
PreOP CTx	−1.41	0.750	0.682	−1.89	**0.044**	[−2.74, −0.02]
Thermoablation	**−2.75**	0.682	0.821	−4.03	**< 0.001**	[−4.40, −1.27]

Predictor	Coefficient	SE (asymptotic)	SE (bootstrap)	z	p	95% CI (bootstrap)
Tumour type 2	−0.09	1.362	1.094	−0.06	0.949	[−2.16, +2.09]
Tumour type 3	+0.65	1.208	1.183	0.54	0.593	[−1.39, +3.23]
Tumour type 4	+0.88	0.920	0.809	0.95	0.341	[−0.61, +2.49]
Tumour type 5	**+5.74**	0.821	0.990	6.99	**< 0.001**	[+4.03, +7.92]
Tumour type 6	−1.36	1.044	0.849	−1.31	0.192	[−2.94, +0.34]
Tumour type 7	**+5.26**	1.205	1.502	4.36	**< 0.001**	[+2.39, +8.25]

Weighted R^2: 0.246

Key Findings

- **Tumour type 5** is the strongest predictor: +5.7 points more discordance than type 1 ($p < 0.001$), robust across all M-estimator norms tested.
- **Tumour type 7** shows +5.3 points more discordance than type 1 ($p < 0.001$), also highly consistent.
- **Thermoablation** is associated with −2.7 points less discordance ($p < 0.001$), the only continuous protective factor.
- **PreOP CTx** reaches borderline significance ($p = 0.044$, asymptotic; bootstrap CI excludes zero barely), associated with −1.4 points less discordance.
- Sex, Age, Zn HIPEC, and tumour types 2, 3, 4, and 6 are **not significant** predictors.

---

4. Model Fit and Residual Diagnostics

Metric	Value
Weighted R^2	0.246
Residual MAD	3.36
Residual RMSE	6.22
Outcome MAD	4.00
Outcome RMSE	6.99
Durbin-Watson	2.014 (no autocorrelation)
Shapiro-Wilk (residuals)	$W = 0.944$, $p < 0.001$ (non-normal)
Skewness	0.94 (right-skewed)
Kurtosis	1.16 (leptokurtic)

---

5. IRLS Weight Distribution

Weight range	N	%
w = 1.0 (not downweighted)	324	78.6%
0.9 w < 1.0	14	3.4%
0.5 w < 0.9	56	13.6%
0.2 w < 0.5	18	4.4%
w < 0.2	0	0.0%

The Huber threshold was 4.52 units; 153 residuals exceeded it. No observations received near-zero weights, indicating the sample contains influential but not extreme outliers.

---

6. Leverage and Influence

- **High-leverage observations** ($h > 2p/n = 0.058$): 26 (6.3%); max leverage = 0.111
- **Cook's D > 4/n**: 22 observations; **Cook's D > 1**: 0 observations; max Cook's D = 0.064

No observation exerts undue influence on the overall fit.

---

7. Sensitivity to M-estimator Norm

Predictor	HuberT (default)	Bisquare (Tukey)	AndrewWave
PreOP CTx	−1.41 (p=0.044)*	−1.56 (p=0.027)*	−1.56 (p=0.026)*
Thermoablation	−2.75 (p<0.001)***	−2.88 (p<0.001)***	−2.88 (p<0.001)***
Tumour type 5	+5.74 (p<0.001)***	+5.10 (p<0.001)***	+5.07 (p<0.001)***
Tumour type 7	+5.26 (p<0.001)***	+5.09 (p<0.001)***	+5.09 (p<0.001)***
Zn HIPEC	+2.54 (p=0.088)	+2.64 (p=0.047)*	+2.65 (p=0.049)*
Scale (MAD)	5.097	4.994	4.990

All significant findings are **consistent across norms**. Zn HIPEC reaches marginal significance (p 0.047–0.049) under Bisquare and AndrewWave norms.

---

8. Standardised Residuals

Threshold	N observations	%
$\|z\| > 2$	42	10.2%
$\|z\| > 3$	14	3.4%
$\|z\| > 4$	3	0.7%
Max $\|z\|$	4.56	—

The proportion of large residuals is slightly elevated relative to a normal distribution, consistent with the right-skewed and leptokurtic outcome.

---

9. Comparison with OLS Results

Finding	OLS	RLM	Conclusion
Tumour type 5	+7.1***	+5.7***	Both significant; RLM attenuated (outlier influence reduced)
Tumour type 7	+5.2***	+5.3***	Consistent
Thermoablation	−2.6***	−2.7***	Consistent
Sex (male)	−1.4*	−0.8 (NS)	OLS effect driven partly by influential observations
PreOP CTx	−0.9 (NS)	−1.4*	RLM reveals effect masked in OLS
R^2 / Weighted R^2	0.224	0.246	Comparable overall fit

The RLM analysis broadly confirms OLS findings but resolves the conflicting signal on Sex (no longer significant) and reveals a clearer effect for PreOP CTx.

---

10. Summary

The RLM model explains ~25% of the weighted variance in sPCI–pPCI discordance. Results are robust to outliers and consistent across multiple M-estimator norms. The dominant predictors are:

1. **Tumour type 5** (+5.7 points vs. type 1, $p < 0.001$) — highest intraoperative surprise
2. **Tumour type 7** (+5.3 points vs. type 1, $p < 0.001$) — second highest discordance
3. **Thermoablation** (−2.7 points, $p < 0.001$) — less discordance, likely reflecting patient selection
4. **PreOP CTx** (−1.4 points, $p = 0.044$) — modest protective association

Sex, age, Zn HIPEC, and tumour types 2, 3, 4, and 6 do not independently predict discordance. The residuals remain non-normal and right-skewed; quantile regression on the median is recommended as a further robustness check.